

NMF for substrate identification in Li-S discharge

Problem statement

- In Li-S batteries, we need to identify the intermediate substrates during the discharge process of the reaction ($16\text{Li} + \text{S}_8 + 16e^- \mapsto 8\text{Li}_2\text{S}$). The intermediate substrates are probed in two ways—UV (UV-Vis spectroelectrochemistry) and Raman spectroscopy (RS). Can we recover the exact proportions of the intermediate quantities?

Data & experimental modalities

- UV probes only the liquid part of the substrate. Not all of the intermediate substrate is in liquid state. The envelope of the total amount of liquid substrate is known. In this modality, the spectra is quantitative both in terms of position, shape, and intensity.
- Raman spectroscopy (RS) measures both liquids and solids. Here, the peak position is quantitative, but the peak intensity isn't. For

Simultaneous data acquisition

- We have simultaneous (currently, "synthetic") acquisition of both modalities during the discharge process, sampled at different frequencies.

Constraints on spectra and loadings.

- *Unimodality of loadings during discharge.* We know that the substrates should obey a unimodal & potentially non-convex & non-concave shape during the discharge process. This shape needs to be imposed on the recovered substrate weights.
- *Spectra of the pure states.* Sometimes the spectra of the pure states is known and could be used.

- *States that exist in liquid state.* It could be assumed that we know roughly which states contribute to the liquid and solid states.
- *Weights at initial part of discharge.* Stefan mentioned that sometimes he could resolve the initial part of the spectra manually, and it could be potentially meaningful if one could solve the problem over a sliding window?

NMF formulation.

Denote $\mathbf{W}_{UV}$, $\mathbf{W}_{RS}$ be the spectra of the pure substrates obtained through UV and RS, respectively, during the discharge process. Denote by $\mathbf{H}_{liq}$ and $\mathbf{H}_{sol}$, the constituents of the substrate in liquid and solid forms, respectively. Furthermore, we have *closedness* (i.e. there is no loss of matter),

$$\mathbf{H}\mathbf{1} = (\mathbf{H}_{liq} + \mathbf{H}_{sol})\mathbf{1} = \mathbf{1}.$$

Now, given spectra from the data of mixture, $\mathbf{E}_{UV}$ and $\mathbf{E}_{RS}$, respectively, recovering the intermediate spectra and weights can be posed as the following optimization problem:

$$\begin{aligned} \min_{\mathbf{W}_{UV}, \mathbf{W}_{RS}, \mathbf{H}_{sol}, \mathbf{H}_{liq}} \quad & \ell_{UV}(\mathbf{E}_{UV}, \mathbf{W}_{UV}\mathbf{H}_{liq}) + \ell_{RS}(\mathbf{E}_{RS}, \mathbf{W}_{RS}(\mathbf{H}_{liq} + \mathbf{H}_{sol})) \\ \text{s.t.} \quad & \mathbf{H}_{liq}, \mathbf{H}_{sol}, \mathbf{W}_{UV}, \mathbf{W}_{RS} \geq 0, \text{ (positivity)}, \\ & \mathbf{H}\mathbf{1} = (\mathbf{H}_{liq} + \mathbf{H}_{sol})\mathbf{1} = \mathbf{1}, \text{ (closedness)} \\ & \mathbf{H}_{liq}\mathbf{1} = \text{liquid profile, (envelope from UV)} \\ & \mathbf{H}_{liq} \text{ and } \mathbf{H}_{sol} \text{ are unimodal.} \end{aligned}$$

In addition to the above constraints, we know the non-zero columns in $\mathbf{H}_{liq}$, we know the spectra of some pure substrates, i.e., we know some rows in $\mathbf{W}_{UV}$ and $\mathbf{W}_{RS}$. The likelihood functions ℓ_{UV} and ℓ_{RS} encode how we choose to impose data fidelity for UV and RS respectively. Since UV is quantitative, we can write

$$\ell_{UV}(\mathbf{E}_{UV}, \mathbf{W}_{UV}\mathbf{H}_{liq}) = \|\mathbf{E}_{UV} - \mathbf{W}_{UV}\mathbf{H}_{liq}\|_2^2.$$

and since RS is qualitative in terms of the peak intensities but quantitative in terms of peak positions, we can demand that,

$$\ell_{RS}(\mathbf{E}_{RS}, \mathbf{W}_{RS}\mathbf{H}) = \|\text{peaks}(\mathbf{E}_{RS}) - \text{peaks}(\mathbf{W}_{RS}(\mathbf{H}_{liq} + \mathbf{H}_{sol}))\|.$$